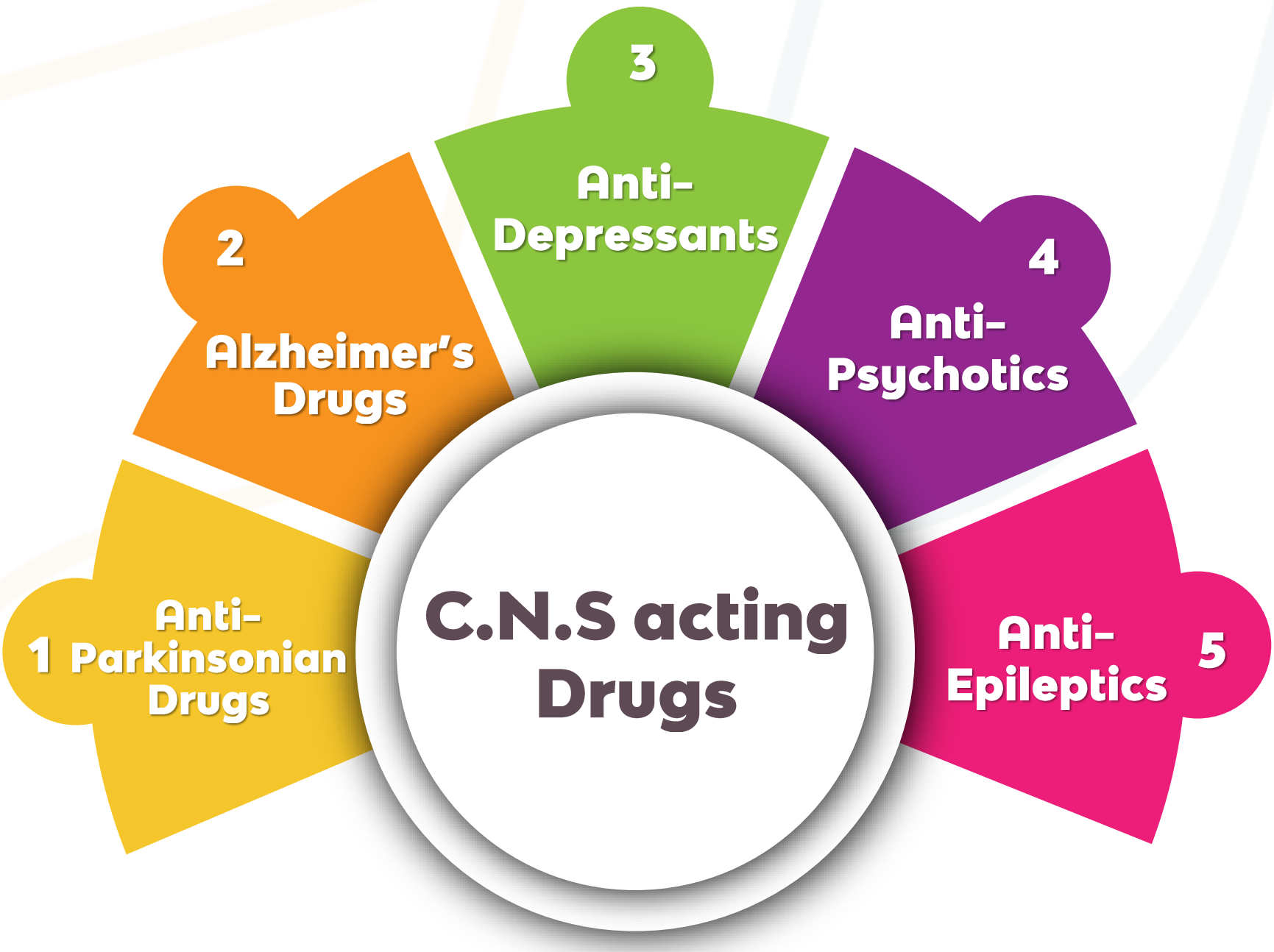
C.N.S



صيدليـة المتحـدة
United Pharmacy

C.N.S drugs classification



Today we will learn about

01 Anti-Parkinsonian Drugs

02 Alzheimer's Drugs

03 Anti-Depressants

04 Anti-Psychotics

05 Anti-Epileptics

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Anti-Parkinson Drugs

Parkinson's disease is a chronic neurological disorder caused by dopamine deficiency in the brain, leading to symptoms like tremors, stiffness, and slow movement. Treatment mainly aims to restore dopamine-acetylcholine balance.

1. Central Dopamine Enhancers ✅
- They Boost dopamine levels in the brain.

2. Dopamine Agonists ✅
- They Mimic dopamine by directly stimulating dopamine receptors.

3. Anticholinergics ✅
- They Block acetylcholine to restore neurotransmitter balance
- This class isn't currently available.

1 Anti-Parkinsonian Drugs

- Central Dopamine Enhancers
  - **SINEMET®** |*Carbidopa/levodopa*
  - **STALEVO®** |*Carbidopa/Levodopa/Entacapone*
  - **PK-MERZ®** |*Amantadine sulfate*
- Dopamine Agonist
  - **PARLODEL®** | *Bromocriptine mesylate*
  - **DOPAXOL** & **SIFROL®** | *Pramipexole*
- Anticholinergics
  - **PARKINOL®** |*Trihexyphenidyl*
  - **COGENTIN ®** |*Benztropine*

Anti-Alzheimer's drugs

Alzheimer's disease is a progressive neurodegenerative disorder causing memory loss, confusion, and cognitive decline. Drug therapy aims to slow symptom progression by modulating neurotransmitters involved in learning and memory.

1. Cholinesterase Inhibitors ✅

- They Increase acetylcholine in the brain to improve neurotransmission and cognitive function.

2. Glutamate Antagonist (NMDA receptor antagonist) ✅

- They Reduce excitotoxicity from glutamate, protects neurons from damage.

2

Alzheimer's Drugs

Cholinesterase Inhibitors

- **DEMENTILE®** | *Donepezil*
- **TRIVACIN & EXELON®** |*Rivastigmine*
- **REMINYL®** |*Galantamine*

Glutamate Antagonist (NMDA)

- **ZEMENTA, MARLON & EBIXA®** |*Memantine*

Anti-Depressant Drugs

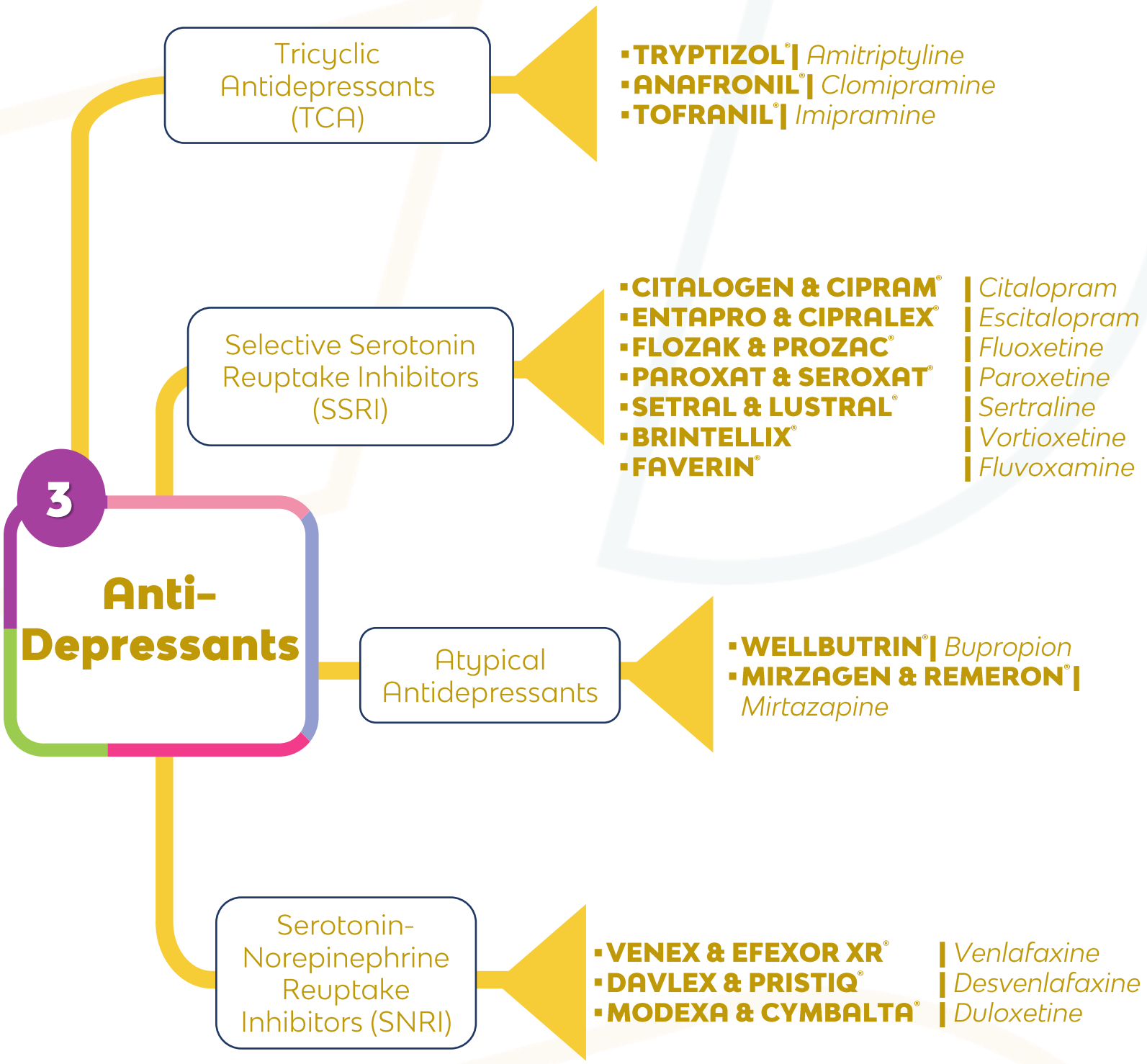
Depression is a common mental health condition characterized by chemical imbalances in the brain, mainly involving serotonin, norepinephrine, and dopamine. Antidepressants help restore this balance.

1. Tricyclic Antidepressants (TCA) ✅
- They Inhibit reuptake of serotonin & norepinephrine.

2. Selective Serotonin Reuptake Inhibitors (SSRI) ✅
- They Increase serotonin by blocking its reuptake.

3. Serotonin-Norepinephrine Reuptake Inhibitors (SNRI) ✅
- They Block reuptake of both serotonin & norepinephrine.

4. Atypical Antidepressants ✅
- They Act via other mechanisms (not just serotonin/norepinephrine).


3
Anti-Depressants
Tricyclic Antidepressants (TCA)
▪ TRYPTIZOL® | Amitriptyline
▪ ANAFRONIL® | Clomipramine
▪ TOFRANIL® | Imipramine
Selective Serotonin Reuptake Inhibitors (SSRI)
▪ CITALOGEN & CIPRAM® | Citalopram
▪ ENTAPRO & CIPRALEX® | Escitalopram
▪ FLOZAK & PROZAC® | Fluoxetine
▪ PAROXAT & SEROXAT® | Paroxetine
▪ SETRAL & LUSTRAL® | Sertraline
▪ BRINTELLIX® | Vortioxetine
▪ FAVERIN® | Fluvoxamine
Atypical Antidepressants
▪ WELLBUTRIN® | Bupropion
▪ MIRZAGEN & REMERON® | Mirtazapine
Serotonin-Norepinephrine Reuptake Inhibitors (SNRI)
▪ VENEX & EFEXOR XR® | Venlafaxine
▪ DAVLEX & PRISTIQ® | Desvenlafaxine
▪ MODEXA & CYMBALTA® | Duloxetine

Anti-Psychotic Drugs

Psychotic disorders like schizophrenia and bipolar disorder involve imbalances in brain chemicals such as dopamine and serotonin.
Antipsychotics help manage hallucinations, delusions, mood swings, and other symptoms.

1. First-Generation Antipsychotics ✅

- They Also called "typical" antipsychotics, Block dopamine D2 receptors. (Not Currently Available)

2. Second-Generation Antipsychotics ✅

- They Also called "atypical" antipsychotics, Block dopamine D2 and serotonin 5-HT2A receptors.
- They also has lower incidence of causing side effects.

3. Mood Stabilizers ✅

- They Used mainly in bipolar disorder to control mood swings.

4 Anti-Psychotics

- First-Generation
  - *Chlorpromazine*
  - *Haloperidol*
  - *Fluphenazine*
- Second-Generation
  - **ADAZIO & SEROQUEL®** | *Quetiapine*
  - **RESPAL & RISPERDAL®** | *Risperidone*
  - **ZOLINDA & ARIPEX®** | *Aripiprazole*
  - **PREXAL & ZYPREXA®** | *Olanzapine*
  - **SILOMES &SOLIAN®** | *Amisulpride*
  - **LEPONEX®** | *Clozapine*
  - **INVEGA®** | *Paliperidone*
- Mood Stabilizers
  - **DEPAKINE®** | *Valproate*
  - **TEGRETOL®** | *Carbamazepine*
  - **LOXOL & LAMICTAL®** | *Lamotrigine*

Anti-Epilepsy

Epilepsy is a neurological disorder characterized by recurrent seizures due to abnormal electrical activity in the brain. Drugs work by stabilizing neurons and reducing excitability.

1. Sodium Channel Modulators ✅
- They Stabilize inactive sodium channels → reduce rapid firing.

2. Calcium Channel Modulators ✅
- They Block T-type calcium channels in thalamic neurons.

3. Synaptic Vesicle 2A (SV2A) Modulators ✅
- They Bind to SV2A protein → reduce neurotransmitter release.

4. Mixed Mechanisms of Action ✅
- They Act via multiple pathways.

5 Anti-Epileptics

Sodium Channel Modulators

- **LOXOL & LAMICTAL®** | *Lamotrigine*
- **CARPAZIO & TRILEPTAL®** | *Oxcarbazepine*
- **VIMPAT®** | *Lacosamide*
- **EPANUTIN®** | *Phenytoin*
- **TEGRETOL®** | *Carbamazepine*

Calcium Channel Modulators

- **CONVENTIN & GAPENTIN & NEURONTIN®** | *Gabapentin*

Synaptic Vesicle 2A Modulators

- **KEPPRA, EPITAM & VALES®** | *Levetiracetam*

Mixed Mechanisms of Action

- **IPRAMAX & TOPAMAX®** | *Topiramate*
- **DEPAKINE®** | *Valproate*

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط